

Nishant Choudhary

B.Tech Electronics & Computer Engineering | VIT Chennai | 2027

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EXPERIENCE

Mimic Productions — AI Programming Intern

July 2026 – Present

Python · Pillow · NumPy · rembg/U²-Net · fal.ai · OpenAI API

Remote

- Built a Python pipeline rendering brand marketing carousels entirely in code — 12 layouts × 4 colour treatments (**48 looks**) at **~1 s/image** and zero marginal cost, lifting a hard ~7-image-per-session ceiling.
- Engineered brand-palette extraction from any reference image (quantisation, perceptual merging, **WCAG** contrast checks), onboarding brands with no documented style guide; validated across **8 visual styles** (photoreal, 3D, anime, stop-motion, 2D).
- Benchmarked image-generation providers (OpenAI **gpt-image-1** vs FLUX via fal.ai) and found **context-gathering calls, not generation, dominated API cost** — redirecting the team from a provider switch to caching, with a substantially larger projected saving.

SELECTED PROJECTS

ClickTutor AI

🐙 GitHub · 📄 Writeup

Python · PyQt6 · Gemini API · Tesseract

- Raised anchor localisation from **79% to 33/33** across 8 screens with a bounded repair loop; classified the failures first and found the 6-pass matcher responsible for **none** of them.
- Caught the locator returning confident wrong boxes — **9 of 24** on hard screens. Which match pass fired predicted correctness exactly (**phrase 40/0, word 0/9**); routed on it, validated **25/26** on 7 held-out screens.
- Benchmarked OCR against vision-model localisation on two corpora — OCR **97%/29%**, vision **76%/96%** on readable/unreadable screens — and routed by screen type instead of replacing either; cost measured at **1.12 API calls per lesson**.

Fake News Detection REST API

🌐 Live Demo · 🐙 GitHub

Python · FastAPI · BERT · ONNX Runtime · Docker · Hugging Face

- Deployed a FastAPI service around a fine-tuned BERT 6-class LIAR classifier (**/health** probe returning **503** when weights are absent, Docker image verified healthy) at **8.5 ms** mean / **12.8 ms** P95 and **396 req/s** batched — **25.7%** accuracy on a dataset-limited 6-way benchmark (16.7% chance), stated up front because the project targets serving engineering, not the metric.
- Benchmarked **ONNX Runtime** export and int8 quantisation on a **separate model-level harness** (torch fp32 **236 ms** P95 → int8 **86 ms**): kept fp32, **rejected int8** despite 2.7× lower P95 — McNemar **p=0.38** left the drop insignificant, but only **57%** of predictions matched, so equivalence was unproven.

Government Budget Analytics Platform

🐙 GitHub

Python · MySQL 8 · Streamlit · pandas

- Cut a **922K-row** (921.5K synthetic) MySQL dashboard's page load **7.2×** (3m37s → 30s); root-caused from EXPLAIN plans to MySQL not pushing disjunctive predicates into a **GROUP BY** view.
- Measured a covering index the docs credit with 80–85% at **0.8%** — inside run-to-run noise — and published the negative result; buffer-pool sizing was the real win at **32.7%**.

TECHNICAL SKILLS

Languages & Data Python, C++, SQL/MySQL (Schema Design, Query Optimization, Indexing, ETL)

AI / ML Deep Learning, Computer Vision, NLP, Transformers, Transfer Learning, Diffusion Models, Image Segmentation, Model Evaluation, Quantisation

Frameworks PyTorch, TensorFlow, Scikit-learn, OpenCV, Hugging Face, ONNX Runtime, FastAPI, Streamlit, Pandas, NumPy

Engineering REST APIs, Docker, Git/GitHub, GitHub Actions CI, Linux, Profiling & Benchmarking, pytest, ruff, mypy

EDUCATION

B.Tech — Electronics & Computer Engineering, VIT Chennai CGPA: **8.32 / 10** Aug 2023 – May 2027

ACHIEVEMENTS & CERTIFICATIONS

- Advanced Generative AI** — IBM Career Education Program (Jul 2026) · **Oracle OCI AI Foundations Associate** (2025) · **Machine Learning Specialization**, Andrew Ng (Coursera) · **150+** DSA problems solved on LeetCode.